

On the appropriate objective function for post-disaster humanitarian logistics models



José Holguín-Veras^{a,b}, Noel Pérez^c, Miguel Jaller^c,
Luk N. Van Wassenhove^{d,*}, Felipe Aros-Vera^c

^a William H. Hart Professor, Rensselaer Polytechnic Institute, Troy, NY, USA

^b Director of the Center for Infrastructure, Transportation, and the Environment Rensselaer Polytechnic Institute, Troy, NY, USA

^c Department of Civil and Environmental Engineering, Rensselaer Polytechnic Institute, Troy, NY, USA

^d Academic Director Humanitarian Research Group, INSEAD, Fontainebleau, France

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ABSTRACT

The paper argues that welfare economic principles must be incorporated in post-disaster humanitarian logistic models to ensure delivery strategies that lead to the greatest good for the greatest number of people. The paper's analyses suggest the use of social costs—the summation of logistic and deprivation costs—as the preferred objective function for post-disaster humanitarian logistic models. The paper defines *deprivation cost* as the economic valuation of the human suffering associated with a lack of access to a good or service. The use of deprivation costs is evaluated with a review of the philosophy and the economic literature to identify proper foundations for their estimation; a comparison of different proxy approaches to consider human suffering (e.g., minimization of penalties or weight factors, penalties for late deliveries, equity constraints, unmet demands) and their implications; and an analysis of the impacts of errors in estimation. In its final sections, the paper conducts numerical experiments to illustrate the comparative impacts of using the proxy approaches suggested in the literature, and concludes with a discussion of key findings.

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1. Introduction

Large disasters and catastrophic events such as the 2004 Indian Ocean Tsunami, 2005 Hurricane Katrina, 2010 Haitian Earthquake, and the 2011 earthquake and tsunami in Japan illustrate how challenging humanitarian logistics are in response to extreme events. The critical supplies (e.g., water, food, medicine) available in the affected areas are either destroyed by the event, or quickly depleted during the immediate response, necessitating the rapid deployment of external assistance to reduce further loss of life (Van Wassenhove, 2006; Holguín-Veras et al., 2012b). The complexities of transporting and distributing critical supplies after a large disaster are only now starting to be understood. Central to this complexity is the socio-technical nature of logistics: a social network of individuals conducting a set of technical activities (e.g., routing, inventory management) over a set of supporting systems (e.g., transportation, communications) (Holguín-Veras et al., 2012b). Because of these inter-linkages, the temporary or permanent

disruption of one or more of these components can bring the entire process to a halt. In large disasters and catastrophes—where the social disruption is extreme, and supporting systems and technical activities are seriously impacted—recovering from the initial impacts requires repairing and reestablishing minimal functionality and connectivity in and among all these components. This seemingly modest objective is extremely difficult to achieve, as the relief effort experiences of these large disasters clearly demonstrated.

In a previous publication (Holguín-Veras et al., 2012b), the authors discussed the main (social and technical) similarities and differences between commercial and humanitarian logistics, and established their place on a continuum, with commercial logistics at one end, and post-disaster humanitarian logistics (PD-HL) at the other. Moreover, Holguín-Veras et al. (2012b) argue that the field of humanitarian logistics is too broad to be neatly characterized by a single set of operational definitions. Humanitarian logistics involves all activities conducted at the different phases of emergency management: mitigation, preparedness, response and recovery (National Governors' Association Center for Policy Research, 1979; FEMA IS-1, 2010). The first two are conducted before a disaster, to reduce the potential economic, social and physical impacts of an event, and include such activities as the pre-positioning of critical supplies, improvement of building codes, and the development of response plans. Immediately after a

* Corresponding author.

E-mail addresses: jhv@rpi.edu (J. Holguín-Veras), perez2n@rpi.edu (N. Pérez), jaller@rpi.edu (M. Jaller), vwl@insead.edu, luk.van-wassenhove@insead.edu (L.N. Van Wassenhove), arosvm@rpi.edu (F. Aros-Vera).

disaster, emergency responders initiate the response phase, which includes search and rescue, and transporting the equipment, personnel and materials required for impact assessment and repairs. This phase is shortly followed by recovery activities, of which there are two distinct sub-phases: short- and long-term recovery (Natural Hazards Research and Applications Information Center, 2005, Sections 2–4). Short-term recovery, or the transitional stage between response and long-term recovery, occurs in the most chaotic and challenging post-disaster environment. At this stage, the management of volunteers and donations, damage and impact assessment, debris removal, distribution of critical supplies, securing of safe locations to host beneficiaries, procurement of materials, and the restoration of critical infrastructure all take place. These activities are performed at a time when needs are extreme, resources scarce, supporting systems impacted, and when such post-disaster behaviors as material convergence are present; in sum, this is what the authors term “post-disaster humanitarian logistics” (PD-HL). At the short-term recovery stage the urgency of shipments is highest, reflecting the degree of human suffering involved. However, activities conducted in long-term recovery, where the objective is to return to normal or improved pre-disaster conditions can be termed “regular humanitarian logistics” (Holguín-Veras et al., 2012d). These activities include the delivery of food to chronic crisis regions, and the delivery of medicines to people in need in developing countries (Chomilier et al., 2003; Van Wassenhove, 2006; Tomasini and Van Wassenhove, 2009). Operations in response to the 2011 Tohoku tsunami in Japan (Holguín-Veras et al., 2012d), or the 2011 tornados that impacted the Mid-West region of the United States, or the 2010 Port-au-Prince earthquake (Holguín-Veras et al., 2012a) are all examples of PD-HL.

Based on the field work conducted by the authors, Holguín-Veras et al. (2012b) identified seven key characteristics that differentiate commercial logistics, regular humanitarian logistics, and PD-HL: objective pursued, origination of the commodity flows, knowledge of demand, decision-making structure, periodicity/volume of logistic activities, state of social fabric and networks, and state of the supporting systems (Holguín-Veras et al., 2012b). The analyses concluded that the objective pursued by the different types of logistic endeavors is a critical difference between them, as is, by extension, the objective functions used in the analytical formulations at the core of their corresponding decision support systems. While in the commercial case, minimization of logistic costs is the primary motivation, in the humanitarian case other factors come in to play, most notably the impacts on human suffering. Commercial logistics emphasize ways to efficiently satisfy demand in conditions where supply equals or exceeds demands, while humanitarian logistics emphasize how to distribute a shortage of critical supplies in a manner that leads to the greatest social good. This emphasis requires that PD-HL models explicitly consider the opportunity costs of delivery actions, and leads to models that are structurally different from those inspired by the commercial logistic paradigm.

The objective function is fundamental to deciding how to harmonize conflicting goals and optimally allocate scarce resources. Maintaining high levels of service at the affected areas could require larger logistical expenditures than are possible. On the other hand, minimizing logistic costs without considering the impacts on the population could negatively affect their welfare. Identifying the best combination of logistic effort and level of service is essential, though possible only if the objective function properly measures both aspects. If the model does not establish an isomorphic relation with the real system, its ability to produce sound recommendations will be compromised. The main purpose of this paper is to define how this balance can be accomplished. The paper argues for incorporating a measure of the suffering experienced by the

beneficiaries, from their lack of access to a good or service. This external effect, not incurred by the organizations conducting PD-HL activities, is a “deprivation cost.”

The paper's main objectives are to: investigate appropriate ways to define the objective function of PD-HL analytical models; identify the characteristics that deprivation cost functions should have; discuss philosophical and economic aspects of relevance to deprivation cost estimation; shed light on the adequacy of proxy ways to consider human suffering; and assess the impacts of using these approaches. In conducting these comparative analyses, the authors decided to narrow down the focus of the work to make it fit within the confines of a single paper. First, the analyses consider stylized versions of the mathematical models to simplify the exposition and the identification of advantages and disadvantages. Second, for similar reasons, the paper only considers deterministic versions of the models as the paper's main focus is on the structure of the objective function. Worthy publications that consider stochastic effects (Barbarosoglu and Arda, 2004; Salmerón and Apte, 2010), are not discussed here. Third, the main focus of the discussion is on relief distribution models for PD-HL operations, where deprivation costs and their consideration, are of utmost importance. However, the insights developed are relevant to all humanitarian logistic models that need to consider the impacts on beneficiaries. Notwithstanding the effects of these simplifications, and the likely existence of pathological examples, the authors are confident on the robustness of the paper's conclusions. The authors believe that this paper—in spite of its contributions and limitations—is only the first step in the long march toward the development of humanitarian logistic models that consider realistic objective functions.

The paper is organized as follows: Section 2 discusses the literature on humanitarian logistics; Section 3 provides an overview of the economic and philosophic considerations concerning the social costs and benefits associated with a delivery strategy; Section 4 develops simplified mathematical formulations to analyze the impacts of alternative ways to consider human suffering; Section 5 uses numerical experiments to illustrate the impacts of using such proxy measures; Section 6 explores the impacts of estimation errors in the deprivation cost functions; and the paper ends with Section 7, a summary of key findings.

2. Literature review

A selective literature review was conducted, with an emphasis on key publications that use proxy measures as their objective functions. For a comprehensive literature review, the reader is referred to (Pérez, 2011). The review was begun with an assessment of the extent to which the existing literature considers the key features of PD-HL operations. On the basis of observations made during disaster research field work (Samii et al., 2002; Van Wassenhove, 2006; Holguín-Veras et al., 2007, 2012a,b,c,d, 2013; Tomasini and Van Wassenhove, 2009; Van Wassenhove and Pedraza-Martinez, 2012), the authors identified the most relevant features that should be incorporated in PD-HL analytical formulations. Moreover, the models were grouped depending on the nature of their objective functions: (1) primarily logistic costs; (2) only a proxy measure of human suffering; and (3) logistic costs and a proxy measure of human suffering. Table 1 summarizes the main features of the models reviewed, which represent the most prominent and widely used approaches.

Inter-temporal externalities, material convergence, and human suffering are essential to consider in proper modeling of PD-HL operations. Inter-temporal effects must be considered because the impacts of delivery actions accrue over subsequent time periods, and cannot be accounted for in single period optimization models. Material convergence is the phenomenon by which large amounts of cargo and equipment, of various degrees of priority, arrive

Table 1
Features of selected humanitarian logistic formulations.

Formulation	Technical features relevant to HL							Objective functions					Other objectives and/or constraints
	Uncertain demands	Dynamic sup-port systems	Variable/ service priority	Multiple decision makers	Inter-temporal externalities	Material conver-gence	Human suffering	MIN Logistic costs	MIN Penalty or Weight Factors	MIN unmet demands	Others	MIN Social cost	
Formulations that primarily consider logistic costs													
Haghani and Oh (1996)								✓					Enforce time windows All demands must be satisfied Stochastic demands All demands must be satisfied
Han et al. (2007)								✓					
Barbarosoglu and Arda (2004)	✓	✓						✓	✓				
Rawls and Turnquist (2010)	✓	✓						✓	✓				
Formulations that consider a proxy measure of human suffering													
Ukkusuri and Yushimito (2008)		✓									✓		MAX reliability of reaching nodes
Yi and Kumar (2007)				✓					✓	✓			
Yi and Özdamar (2007)				✓					✓	✓			
Özdamar et al. (2004)	(*)	(*)								✓			MAX throughput, MIN delivery MIN headway
De Angelis et al. (2007)	(*)	(*)									✓		
Sheu (2007)	✓							✓	✓		✓		
MAX fill rate													
Fiedrich et al. (2000)	✓	✓	✓								✓		MIN number of fatalities MIN expected travel distance. MIN MAX and MIN average arrival time
Chang et al. (2007)	✓	✓		✓				✓	✓		✓		
Campbell et al. (2008)											✓		
Formulations that consider logistic costs and a proxy measure of human suffering													
Salmerón and Apte (2010)	✓			✓				✓		✓	✓		MIN casualties, Budget constraint MIN response makespan MIN travel, MAX MIN satisfaction All demands to be satisfied MIN urgency measure Proxy measures for human suffering Social cost: logistic + deprivation cost
Barbarosoglu et al. (2002)				✓				✓			✓		
Tzeng et al. (2007)	✓	✓						✓			✓		
Balcik et al. (2008)			✓					✓	✓	✓			
Yushimito et al. (2012)											✓		
Jaller (2011)		✓	✓	✓	✓	✓	✓	✓			✓	✓	
Pérez (2011)					✓		✓	✓				✓	
Number of cases	9	9	5	9	2	1	2	12	7	5	11	2	

Note: (*) Propose solving deterministic versions of the problem as new information becomes available. The aspects considered to be crucial to PD-HL are shaded.

at a disaster site (Fritz and Mathewson, 1957). As discussed in Holguín-Veras et al. (2012c) and Holguín-Veras et al. (2013), the non-priority component of this flow produces major complications for the disaster response (Fritz and Mathewson, 1957; Holguín-Veras et al., 2012b), to the point that it is considered a “...second tier disaster...” (Newsweek, 2002). Finally, the authors contend that human suffering is not properly captured by the proxy measures in use. This point is amply discussed in Section 4.

Only two of the models (Jaller, 2011; Pérez, 2011) formally consider these three essential aspects of PD-HL, and the majority consider only some of the technical features listed in Table 1. This undoubtedly reflects the models' commercial logistic roots, and the fact that PD-HL modeling is still in early stages of development. In terms of objective functions, the predominant one is minimization of (variants of) logistic costs, followed by minimization of penalties or weighting factors, while minimization of unmet demands is a distant third. It is revealing that most models require the use of “equity constraints” to ensure that they produce reasonable results. The most frequently used are: maximum headways between deliveries, minimum delivery amounts, and maximum fill rates. The use of such constraints and proxy measures suggests that the model developers recognize the need to consider human suffering, though they try to do so in a simplified manner that does not compromise computational efficiency. These techniques are referred to in this paper as “proxy approaches.”

While relying solely on logistic costs is appropriate for commercial activities, it is certainly not for modeling PD-HL operations. In extreme situations, when demands greatly exceed supplies, the primary goal is to allocate the limited resources to mitigate human suffering. Approaches that simply minimize unmet demands, or maximize the cargo delivered, do not take into account the time that a population may have been without supplies. Such methods do not account for the urgency with which supplies may be needed at different locations, and cannot identify the optimal allocation of those resources to achieve the highest social benefit. Models that use priority or weighing factors may provide more balanced levels of service, though they require subjective definitions of those prioritization or weighing schemes. As such, the solutions achieved are directly correlated to the parameter values used in the optimization; they do not capture the dynamic nature of human needs and suffering. The few formulations considering both operational costs and a proxy measure of human suffering have difficulties reconciling these two objectives. Without the ability to express operational and human suffering in a common scale, these methods rely on subjective parameters that potentially bias the results towards the modelers' expectations.

The modelers who add equity constraints to ensure that their models produce reasonable results undoubtedly understand the inherent limitations. However, such a decision introduces other problems, as equity constraints presuppose the existence of feasible solutions. Since feasibility depends on a balance between the supply delivered by the PD-HL system and the demands to be satisfied, the imposition of equity constraints without a clear idea to what the PD-HL system could deliver, in terms of supplies distributed, leads to infeasible solutions. As a result, modelers are forced to readjust the constraints until they lead to “reasonable” results. As further discussed in Section 4, equity constraints are, by nature, arbitrary. The only way to overcome these problems is to redefine the objective functions used in humanitarian logistic models with the assistance of welfare economics.

3. Social costs = logistic + deprivation costs

The critical social impacts of humanitarian logistics suggest that the corresponding analytical formulations be based on welfare

economics, which is the branch of economics that studies how the allocation of resources affects economic welfare (Varian, 1992). In general terms, economic welfare (the economic wellbeing of the system under analysis) has two primary components: the private costs and benefits accrued by the participants in the economic transactions, and the external costs associated with the impacts that these transactions produce on economic agents not directly involved. Social costs and benefits are the summation of all private and external costs, and benefits, respectively.

These concepts are of great relevance to humanitarian logistics. A crucial observation is that the need for a relief effort arises because an external agent (e.g., an earthquake, a prolonged war or political crisis) imperils or destroys the markets that in normal times allow sellers and buyers to transact for supplies. The ensuing impacts on people—the lack of available currency and credit, the partial or total destruction of infrastructure and markets, and the interruption of private sector supply chains—leaves the provision of humanitarian relief as the only alternative to loss of lives and suffering. The sudden disappearance of the market, and the urgent need to distribute relief aid at no costs to beneficiaries, completely changes the nature of the economic activities involved. In normal conditions, buyer-seller transactions are private, with costs and benefits internalized by the participants, and welfare is well defined by the summation of consumer and producer surpluses (Varian, 1992). After a disaster destroys the market, no such internalization takes place because normal economic transactions are impossible. With the onset of the disaster, the economic impacts of the distribution of supplies—which in normal conditions are captured by producer and consumer surpluses—become externalities.

Three main agents must be considered to properly compute economic welfare, at any given point in PD-HL: the relief group, the beneficiaries of the aid, and the individuals who do not receive aid at that time. While other economic agents could be impacted, e.g., local producers that experience major losses if a large inflow of aid depresses local markets, the main focus here is on the short-term response, so these important medium-term effects are disregarded. In terms of economic welfare, the relief agency incurs the logistic costs associated with the purchase, transportation, inventory, and distribution of relief supplies. The most important aspects to consider are those associated with the distribution of that aid. Aid distribution produces two radically different effects: a reduction of deprivation costs for the beneficiaries; and, frequently overlooked, an increase in deprivation costs for those individuals who did not receive relief aid at that time. Because of the lack of a market, these impacts become externalities: the reduction of deprivation costs to beneficiaries is a positive externality, while the increased deprivation cost for others is a negative externality. The latter is the “opportunity cost” of the relief effort.

These components of economic welfare have different characteristics in terms of time-dimensionality. Logistic costs and reductions of deprivation costs typically take place in a single time period (except if a large delivery is made to a demand node, and the supplies are saved for later use). The same cannot be said about the opportunity costs for those individuals who did not receive aid; these are likely to accrue in future time periods. Not delivering to a set of demand nodes is almost unavoidable because of the scarcity of supplies, and it will increase deprivation costs, as the population at those nodes will have to wait for the next delivery epoch to receive aid. For PD-HL models to properly account for the full impacts of delivery decisions, they must assess these inter-temporal externalities with multi-period optimization models. Single period formulations cannot correctly account for opportunity costs and should be avoided in the PD-HL context.

PD-HL models should be reformulated to explicitly take into account the logistic costs, as well as the impacts of the deliveries, which are the inter-temporal opportunity costs and benefits. This

would provide a solid foundation for decision makers to design and plan an effective use of scarce resources. However, using a formal metric to value deprivation costs leads to philosophical and ethical dilemmas: Is it ethical to aggregate individual welfare metrics? Is it ethical to purposely leave someone in peril? The rest of this section discusses these important questions from both philosophical and economic viewpoints.

3.1. Philosophical considerations

Using economic efficiency as a metric for resource allocation in crisis situations is controversial because of its ethical and moral implications. One school of thought argues that society has a moral obligation to help and rescue individuals identified as being in immediate peril, even if doing so means high costs or economic inefficiencies. This perspective, known as the “Rule of Rescue,” is based on the deontological theory of ethics, which states that the morality of an action is determined by its adherence to prescriptive rules, rather than by its consequences (Pinkerton et al., 2002). Actions are morally acceptable as long as they fulfill an expected “duty.” Deontological ethics stand in contrast to consequentialist ethics, which state that acts are right or wrong on the basis of their consequences, implying a duty to produce the best result (Kamm, 1993). Consequentialists focus on (social) cost effectiveness, arguing that decision makers should allocate scarce resources in ways that render as much good as possible. A prominent example of this school of thought is utilitarianism.

Few would argue against using the Rule of Rescue when individuals are at serious risk, and there are sufficient resources to meet all needs. Thus, “. . . people cannot stand by idly when an identified person’s life is visibly threatened if effective measures are available. . .” (Hadorn, 1991). In fact, societies that do not accommodate such departures from cost effectiveness may be considered inhumane (Cookson et al., 2008). The Rule of Rescue, however, is of little guidance when resources are scarce, as it does not help with the difficult issue of how to allocate those finite social resources. In fact, applying the Rule of Rescue in such situations poses major difficulties to allocation strategies seeking effectiveness (Pinkerton et al., 2002). After evaluating the use of the Rule of Rescue in Australia and the United Kingdom, Cookson et al. (2008) conclude that, although policy makers should facilitate exceptional departures from the cost effectiveness norm, they should not necessarily implement policies that exonerate certain groups of individuals—including those in imminent danger of death—from taking into account the opportunity costs of their actions.

Recognizing the limitations of the Rule of Rescue, followers of the consequentialist school hold that the criteria for social decision-making should be fundamentally individualistic and utilitarian, and that policies and actions should reflect the preferences of the individual members of society (Mehrez and Gafni, 1990). Although the foundations of utilitarianism date back at least to the Greek philosopher Epicurus (Pinkerton et al., 2002), its cornerstone—the “greatest happiness principle”—was first discussed by philosophers J. Bentham in 1789 and J. Stuart Mill in 1863 (Bentham, 1789, 1948; Mill, 1863, 1987). This principle basically states that “. . . actions should be judged by their consequences and that actions are right or good insofar as they produce the greatest benefit among all affected. . .” (Moskop and Iserson, 2007). Utilitarianism’s fundamental tenet is that, when considering the consequences of one’s actions, everybody’s interests must be taken into account. While the outcome is not necessarily equal for each person, the goal is to produce the greatest benefit. Utilitarians believe that the goal of moral conduct is to produce maximal happiness. They also believe that the resulting utility can be measured, making it suitable for mathematical analysis evaluating the desirability of actions (Pinkerton et al., 2002). The principle of utility may offer, therefore,

some moral justification for the aggregation of costs and benefits when deciding how “best” to distribute critical supplies.

However, even the greatest utility principle leads to difficult questions concerning the proper prediction of consequences, the creation of unequal outcomes (Moskop and Iserson, 2007), and the aggregation of individual welfare. Deciding how best to distribute critical supplies on the basis of the expected consequences of those helped; and what, how, and how much to distribute; requires that those consequences be fairly and accurately predicted. The obvious problem in crisis situations is that accurate predictions are impossible, given the uncertainty and changeability of needs and capabilities. Another important issue is the creation of unequal outcomes; a direct consequence of scarce resources and the impossibility of satisfying everyone’s needs. Although from the utilitarian perspective, such unequal outcomes are acceptable, as long as they lead to the “greatest benefit,” the implied aggregation of welfare has been hotly debated. The principle of equal chances (Taurek, 1977) rejects the notion that individual welfare can be aggregated. “Suffering is not additive in this way. The discomfort of each of a large number of individuals experiencing a minor headache does not add up to anyone’s experiencing a migraine. In such a trade-off situation as this we are to compare (a person’s) pain or loss, not to (the) collective or total pain, whatever exactly that is supposed to be, but to what will be suffered or lost by any single one of (the individuals)” (Taurek, 1977). Taurek states that in a life and death situation where a decision must be made about whom to save: either one person or a group of people, maybe a coin should be flipped, or maybe the same amount of resources should be distributed to each individual. Taurek argues that since each person’s life is equally valuable to him or her, everybody should be given the same chance of survival, as doing so best expresses an equal concern and respect for each person (Moskop and Iserson, 2007).

Other scholars have argued against Taurek’s view that the number of lives saved is morally irrelevant. Sanders (1988) states that, even though he agrees with Taurek’s view concerning the aggregation of individuals’ welfare, the issue is not that there is a greater loss to the total lives, but rather that more people suffer an equal loss, therefore the numbers do count. Sanders also argues that Taurek’s principle of equal chances, where the numbers do not count, rules out any consideration that people are worth saving just because they are people. Kamm (1985) acknowledges the importance of respect for the worth of each individual person. However, she argues that in the event of extremely limited resources all lives cannot be saved, concluding also that emergency responders should honor as many requests for live-saving care as possible (Moskop and Iserson, 2007).

Humanitarian logistics is not the only human activity struggling with these complex issues. Similar questions have been raised in relation to rationing schemes for organ transplantation (Annas, 1985), health policy (Morrow and Bryant, 1995), value of life (Zelizer, 1978), and many other spheres, without reaching consensus. However, medical triage systems provide important lessons, as their purpose is to achieve a fair and efficient use of scarce medical resources (Moskop and Iserson, 2007). Triage is routinely used in mass casualty disasters, where decisions have to be made concerning who will and who will not receive treatment. The Haiti earthquake, quite tragically, provided plenty of examples of medical triage (Associated Press, 2010). As with the other aspects discussed here, there is no clear guidance on the most appropriate mechanism to allocate resources in disaster response. However, the authors’ position is that such a lack of a universally accepted path should not impede or preclude formal consideration of human suffering in PD-HL formulations. The use of social cost as the objective function leads to delivery strategies that work toward equal outcomes to all population segments, where other objective functions do not. This difference, which will be made clear later in the paper,

provides some assurance that the concept outlined in this paper is on the right track.

3.2. The role of economic valuation

The assumption that human suffering could be represented by a “deprivation cost” leads to the practical question of how it should be valued. The lack of a proper market requires that deprivation costs be valued using economic valuation techniques, which, as discussed in Bateman et al. (2002), is the process of assigning economic value to non-market goods and services by measuring the change in wellbeing associated with their provision. These benefits and costs are directly related; the measure of any benefit is that cost which would exactly offset it, and the measure of any cost is that benefit which would offset it. The aim is to see how people respond to a range of choices, and thus to establish the extent of their willingness to pay for a particular benefit, or their willingness to accept payment in exchange for bearing a particular loss. Since wellbeing reflects what individuals prefer, there is a reasonable connection between preferences and willingness to pay. Thus, the latter could be understood as an indication of preference satisfaction that, in turn, could be an indication of wellbeing. Since this type of valuation defines benefits and costs in terms of an individual's own preferences, there is no need to make subjective decisions about what is good or appropriate for them. Therefore, deprivation costs could be measured and valued as changes in wellbeing.

Overall, these estimates of benefits and costs are based on concepts of economic efficiency. When reallocating resources, economic efficiency increases when the benefits to those who gain surpass the losers' costs, or, when the winners' total willingness to pay exceeds the losers' total willingness to accept the losses. Some critics consider economic valuation based on these notions morally questionable, arguing that budget constraints should not be considered when evaluating the delivery of critical supplies to disaster sites. Here again, the authors' position is that the potential benefits of using an unbiased, consistent—though imperfect—metric of human suffering outweighs the negatives of using simpler approaches that provide only crude approximations of the problem's full complexity.

3.3. Conceptual discussion of deprivation costs

Conceptually, deprivation costs are the economic value of the human suffering caused by a lack of access to a good or service. As such, the deprivation cost is likely to be a function of the deprivation time and the socio-economic characteristics of the individual (e.g., age, gender, physical condition). The deprivation cost is expected to be: (1) monotonic, non-linear, and convex with respect to the deprivation time; and (2) associated with non-additive demands and, possibly, hysteretic effects that reflect residual damage to the beneficiaries. Monotonicity is expected, as increases in the deprivation time are bound to lead to increases in deprivation costs across the entire range of values. Convexity is anticipated, because deprivation costs are expected to grow increasingly faster as the deprivation time increases. Non-linearity is a natural consequence of how human beings deal with shortages of life-sustaining items. At first, most healthy individuals are able to deal with short-term shortages without much problem, as they use up their body reserves to operate in a normal or quasi-normal fashion. In the non-hysteretic range, the deprivation cost imposed on the individual by the crisis dissipates once the need is fulfilled. In contrast, if the deprivation enters the hysteretic range—as body reserves are exhausted—body functions slow and start to shut down, and a point may be reached where damage is irreversible and death occurs. In such a case, the cycle of deprivation and needs fulfillment creates

residual impacts (costs) that cannot be eliminated (Cross, 1993). These are the lessons of the classic study of Keys et al. (1950), the “Minnesota Semi-Starvation Experiment,” that used volunteer human subjects to gain insight into the physical and psychological effects of semi-starvation, and how best to feed civilians who had been starved during World War II (Keys et al., 1950; Kalm and Semba, 2005; Alberda et al., 2006).

Formally, a hysteretic economic process is one that retains a memory of past processes (shocks), a memory that is non-linear, selective, and with remanence (Cross, 1993; Hughes-Hallett and Piscitelli, 2002). The memory's non-linearity indicates that its magnitude is non-proportional to the impacting shock. The selectivity property indicates that not all shocks are remembered, only the non-dominated one. Remanence indicates that a reversal of the shock does not take the system back to its starting point, as there are residual effects that persist over time. There is a long tradition of research on economic hysteresis. Alfred Marshall indicated that “... if the normal production of a commodity increases and afterwards again diminishes to its old amount, the demand price and supply price are not likely to return, as the pure theory suggests they will, to their old positions for that amount. ...” (Marshall, 1890). However, economic hysteresis would only come into vogue in the 1980s (Cross, 1993) when economists used it to explain trends in unemployment rates (Blanchard and Summers, 1986; Darity and Goldsmith, 1993; Jacobson et al., 1997), and more recently, the effects of unemployment on wellbeing (Cole et al., 2009).

Consider the case in Fig. 1, in which a deprivation cost function $\gamma(\theta, \delta_{it}, Z)$ for a node i is depicted as a function of time t (where, θ is a parameter vector, δ_{it} is deprivation time, and Z are the individual's socio-economic attributes). Panel (a) shows the case in which there is no hysteresis. As illustrated by the upward and downward arrows, an individual who is deprived of a good would experience the increasing deprivation costs represented by the path AB. Once the needs start to be satisfied, the deprivation costs decline (along BA). At the end, the individual is in the same condition as before the shock. In the hysteretic case in Panel (b), the cycle of deprivation and needs fulfillment do not lead to the elimination of the deprivation costs, as the deprivation cycle's path (AB) is different from the one followed during the fulfillment cycle (BC). Thus, there are residual effects as the original cost function $\gamma(\theta, \delta_{it}, Z)$ is replaced with a new one $\gamma'(\theta', \delta_{it}, Z)$ (in dashed lines) that captures the new state of the impacted system. Should a new deprivation cycle occur, the deprivation cost would follow the path CB, and may lead to a new deprivation cost function $\gamma''(\theta'', \delta_{it}, Z)$ (not shown in the figure).

The non-additive nature of unmet demands is another complication. An individual who has been deprived of a good or service c that is consumed at a rate of u_c per unit time, for a deprivation time of δ_{it} time units, may not be able to consume the total of $u_c \delta_{it}$ when the needs are fulfilled. The case of emaciated prisoners during World War II who died after eating a normal meal is a dramatic example of this phenomenon. In contrast, a business that is deprived of an input for a period of time is likely to be able to use it once the input is available again (unless the absence of the input lead to a loss of customers or to another non-recoverable impact).

There are two main inter-temporal externalities, which must also be considered. The first takes place when a large delivery is made to a node. A portion of the supplies is used to satisfy immediate needs, and the rest could be saved for later use, impacting deprivation costs in future time periods. The second, and probably most common case in the initial stages of the response, is when demand nodes that do not receive aid at a given time period experience a concomitant increase in their deprivation costs, or, the opportunity costs of the delivery strategy. Far from being a theoretical curiosity, these opportunity costs are very real consequences of scarce resources.

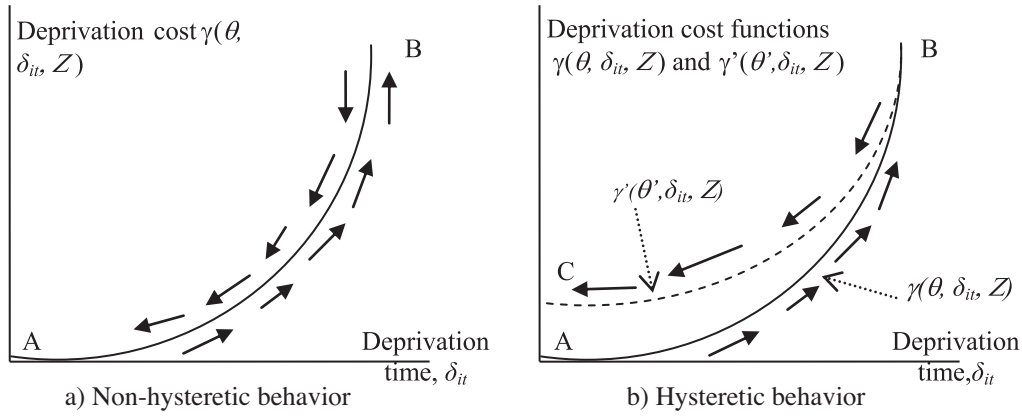


Fig. 1. Schematic of hysteretic and non-hysteretic behavior. Note: θ is a parameter vector, δ_{it} is deprivation time, and Z are the individual's socio-economic attributes.

3.4. Applicability to other contexts

The concepts of deprivation and social costs described in this paper can be used to analyze a number of emergency response functions, such as search and rescue and medical assistance, where the primary intent is to mitigate human suffering. Although in some of these cases, the deprivation costs are a function of the lack of service instead of a lack of critical supplies (which may require estimation of different deprivation cost functions), the underlying concepts are still the same. Similarly, the use of social costs as the objective function is appropriate for a variety of decision contexts. This may include long-term relief cycles, such as those needed in regions suffering prolonged humanitarian crises where chronic malnutrition may lead to permanent damage and the need to consider hysteretic costs, or short-term relief cycles like those typical of disaster response which is the case discussed in the paper. The use of social costs requires no assumptions in terms of the structure of the distribution network. The only related assumption in this regard is the existence of a relief group that is in charge of distributing critical supplies to a population in need that receives no supplies from other groups. This case is typical of PD-HL operations. Essentially, the use of social costs as the objective function is appropriate for cases where the decision process must consider logistic costs and the impacts (externalities) of the provision of services or supplies on a set of beneficiaries.

4. Mathematical modeling with social costs

To analyze the implications of using proxy measures of human suffering in PD-HL modeling, the authors studied the impacts of alternative proxies using simplified analytical formulations representing key model types. Only models developed with post-disaster response as the intended application were considered. These simplified models provide a rather compact analysis that clarifies the advantages and disadvantages associated with the use of the various PD-HL models, without requiring complex formulations that could obscure the interpretation of results.

The formulations consider a node i at which π_{it} observationally identical individuals experience the same deprivation cost. This assumption implies that an outside observer would value the individuals' suffering equally, and individual socio-economic characteristics are not a factor. As a result, the original deprivation cost function $\gamma(\theta, \delta_{it}, Z)$ can be replaced by its generic counterpart $\gamma_g(\theta_g, \delta_{it})$. Although this decision could be contested on philosophical grounds, in most PD-HL operations there are no detailed data about the people in need, so using a generic function may be the only option. The models considered in this section assume: that there

are no residual effects after a deprivation period, i.e., the costs are non-hysteretic; that partial or total fulfillment of needs reduces the deprivation costs at the demand node where the delivery was made; and, that there could be an arbitrary number of demand nodes.

The formulations consider a relief group that delivers critical supplies to a set of demand nodes $i \in N$. The key relief group's decisions are in what order, how much, and when to deliver to the demand nodes in need of supplies, during a planning horizon T . These decisions could be represented as an ordered sequence $X = \{(i_1, d_1, t_1), (i_2, d_2, t_2), \dots, (i_j, d_j, t_j), \dots, (i_J, d_J, t_J)\}$, where the subscript j represents the position in the sequence X (for simplicity the figures omit the subindex j); J is the cardinality of the sequence, $J = |X|$; i_j identifies the node visited during delivery j ; d_j is the amount of supplies delivered at i_j ; and t_j is the time at which delivery j is made. Since the issue of how to determine X is of secondary relevance to the purposes of this paper, X is assumed to be feasible and obtainable through the use of heuristics or algorithms. The formulations assume that two mathematical operators $\Omega_T(X, T)$ and $\Gamma_T^*(X, T)$ compute the total logistic and deprivation costs, respectively, associated with X during the planning period T . In the case of $\Omega_T(X, T)$, it may include, among others, vehicle routing, inventory, and order processing.

4.1. Social cost formulation

The objective function based on social cost includes all private (logistic) costs plus the impacts of the delivery actions on the population (i.e., reductions of deprivation costs to the recipients of the aid, and increases of deprivation costs for the rest of the population). The deprivation cost at node i at time t is equal to the aggregation of individual costs:

$$\Gamma_i(X, t) = \gamma_g(\theta_g, \delta_{it})\pi_{it} \quad \forall i \in N \quad (1)$$

where, $\Gamma_i(X, t)$, deprivation costs incurred at node i at time t as a result of implementing the sequence of delivery actions X ; $\gamma_g(\theta_g, \delta_{it})$, generic deprivation cost function with parameter vector θ_g ; δ_{it} , deprivation time for node i at time t ; π_{it} , population in need at node i at time t ; t , continuous time index, $0 \leq t \leq T$.

The correct computation of total deprivation costs requires proper accounting for the inter-temporal externalities, which requires that the impacts of delivery decisions on deprivation costs be well understood. Fig. 2 is a time dependent depiction of how the deprivation costs evolve in response to delivery actions to a single demand node i (Pérez, 2011). The discontinuous cost function $\Gamma_i(X, t)$ in the figure—capturing the deprivation costs at the node at any time t —shows the deprivation costs increase with

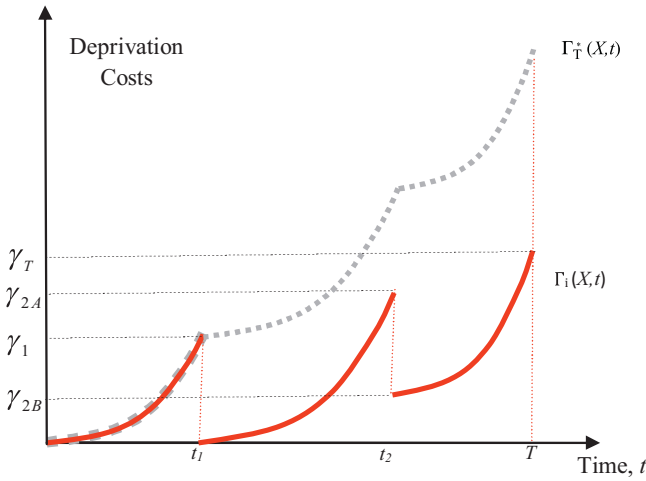


Fig. 2. Impacts of delivery actions on deprivation costs.

deprivation time, and then drop to zero when a delivery large enough to fulfill the needs of all individuals is received at time t_1 . However, by that time, beneficiaries had already experienced a deprivation cost γ_1 that enters into the computation of the total deprivation cost $\Gamma_T^*(X, t)$. Note that at the time of the delivery, the deprivation time for the beneficiaries resets to zero (once they consumed the supplies), while the continuous time index does not. The situation at time t_2 illustrates what happens when the supplies delivered are less than what is required to fulfill the needs of all the individuals in the node, under the assumption that some individuals had their needs satisfied while the rest of the individuals did not. The deprivation costs reset to zero for beneficiaries receiving the aid, and continue to increase for those who did not. The net effect is that deprivation costs at the node decrease only by $(\gamma_{2A} - \gamma_{2B})$. Decisions concerning future deliveries must consider the needs of these two population segments independently.

Since the objective of the PD-HL formulations is to identify the delivery decisions that would minimize total social costs over the planning horizon T , the inter-temporal externalities must be computed for all demand nodes. First, among the notable computational challenges is that $\Gamma_T^*(X, t)$ is non-convex. Second, deprivation times are also affected by the size of the deliveries, as supplies could be saved for later use. This leads to a situation in which the tracking of inventory levels, consumption, deliveries and vehicle trips—necessary to correctly assess the impacts of the distribution strategies—requires a large number of binary variables, and becomes computationally prohibitive for large problems. Fortunately, the structure of the problem allows for exploiting its features to reduce the computational effort: the total deprivation costs up to time T can be determined by evaluating the cumulative cost function at that time epoch, as $\gamma_1 + \gamma_{2A} + \gamma_T - \gamma_{2B}$. Thus, the inter-temporal opportunity cost for the demand nodes that did not receive supplies at a given time t are included in the objective function as accruing at future delivery epochs. To properly consider the inter-temporal externalities, the objective function models must also include the increase in deprivation costs that takes place between the last delivery to each node and the end of the planning horizon T . These steps will ensure that all inter-temporal costs are accounted for, while failing to consider these terminal costs will inevitably lead to erroneous results. In sum, the total social cost is equal to the summation of the deprivation costs at the discrete time epochs at which deliveries were made, plus the summation of the deprivation costs for all nodes outstanding at the end of the planning horizon T .

In mathematical terms, the total deprivation costs up to time t can be expressed as:

$$\Gamma_T^*(X, t) = \sum_{j=1..j': t_j' < t} \Gamma_{ij}(X, t_j) + \sum_{i=1}^N \Gamma_i(X, t) \quad (2)$$

For the entire sequence, X , that takes place over the planning horizon, the total deprivation cost is:

$$\Gamma_T^*(X, T) = \sum_{j=1}^J \Gamma_{ij}(X, t_j) + \sum_{i=1}^N \Gamma_i(X, T) \quad (3)$$

Thus, the total social cost for sequence X is:

$$\Psi_T^{SC}(X, T) = \Omega_T(X, T) + \Gamma_T^*(X, T) \quad (4)$$

$$\Psi_T^{SC}(X, T) = \Omega_T(X, T) + \sum_{j=1}^J \Gamma_{ij}(X, t_j) + \sum_{i=1}^N \Gamma_i(X, T) \quad (5)$$

Eq. (5) expresses total social costs as the summation of the logistic costs incurred in executing strategy X and the inter-temporal effects. Conceptually, the second and third terms capture the opportunity costs of the delivery strategy: the second term evaluates deprivation costs at the delivery epochs; the third term computes the terminal deprivation costs outstanding at time $t = T$ (see Fig. 2) at all nodes in the system (including those not served by X). Explicitly considering all these components is essential for proper modeling of PD-HL. A unique and important feature of Eq. (5) is that it need not impose any constraint on the level of service to be provided by the PD-HL system, as the level of service is endogenously determined. In cases where the PD-HL system is able to deliver the critical supplies needed, the optimal deprivation costs would be relatively low, but if demands greatly exceed the PD-HL system's capacity, deprivation costs will be much larger.

The rest of the paper discusses the models based on a commercial logistic formulation modified to account for the humanitarian goal. The analyses focus on selected model types, to facilitate comparisons, though selecting the model types presented some challenges. Given that most models have common features, there are no clear differentiations. A second challenge is that different models use different variables. While variables such as “time between deliveries” could map well into deprivation time (the key variable in the social cost model), others, such as “fill rate,” are much less useful, and general conclusions about the worthiness of some variables are impossible. After pondering how best to proceed, the authors decided to classify current PD-HL models on the basis of how well their mathematical structures consider human suffering, assuming that the models use deprivation time as their key variable. This decision is conservative in the sense that it provides the models with the best chance to perform well against the social cost model. Ultimately, the authors decided to focus on three key instances of penalty-based models (hard constraint, constant penalty, and variable penalty models), as well as the unmet demand model.

4.2. Penalty-based formulations

The family of penalty-based formulations is rooted in commercial logistic models that are modified with the addition of penalties. The range of modifications is fairly wide, including the addition of hard constraints (a form of penalty) to what otherwise would be a commercial logistic model, and the use of objective functions that consider variable penalties as a function of the tardiness of deliveries. Not surprisingly, the quality of the delivery strategies produced is also wide. In some cases, the penalties are too coarse to capture deprivation costs in any meaningful way; while in others

the penalties could indeed replicate deprivation costs with relative accuracy. The model needs to adequately represent deprivation cost functions, but it also, importantly, has to correctly capture the opportunity costs associated with the delivery strategy. Penalty-based models ignore the increase in human suffering that accrues from the last delivery until the end of the planning horizon. Considering terminal costs is crucial for proper modeling because, in the humanitarian logistic case, once a delivery is made and the supplies are used up, the deprivation costs start rising again. Not considering these terminal costs leads to a model that is underspecified and produces trivial solutions (in the absence of other constraints, the summation of logistic and deprivation costs is minimized by not delivering to anyone). Analysts then add “equity constraints” to enhance the quality of the solutions, which in turn generates other problems, discussed later in this section. The family of penalty-based models includes a variety of alternative forms. Structurally, the models could be classified as based on: hard constraints, constant penalties, and variable penalties. For illustration purposes and to facilitate exposition, the figures in this section only show the first delivery to a single node (assuming no initial inventory), and express deprivation costs as a function of deprivation time.

4.2.1. Variable penalty models

This class of models relies on variable penalties that are a function of the tardiness of a delivery, which are incorporated in the objective function, in a manner similar to vehicle routing models with soft time windows. Thus, late deliveries are allowed, though they are penalized for not meeting a preset threshold δ_p . In most cases, the variable penalties are specified as piece-wise linear functions of the difference between the solution and the pre-defined threshold $\delta'_{it_j} - \delta_p$, where δ'_{it_j} represents the deprivation time outstanding at the time of delivery and $\delta'_{it_j} > \delta_p$; and ρ' is the unit penalty. For the model with a two single piece-wise linear component and a penalty equal to zero if $\delta'_{it_j} < \delta_p$ (terms representing additional piece-wise linear segments could be added if so desired), the objective function can be formulated as:

$$\Psi_T^{VP}(X, T) = \Omega_T(X, T) + \sum_{i=1}^N \sum_{j=1}^J \rho'(\delta'_{it_j} - \delta_p) y'_{it_j} \quad (6)$$

Where: y'_{it_j} is equal to 1 if $\delta'_{it_j} > \delta_p$, and 0 otherwise.

As shown, Eq. (6) computes the penalties accrued at all nodes i for which the delivery time is larger than the threshold when evaluating at every delivery time t_j . Structurally, the variable penalties depend on the magnitude of the violation, as opposed to those that are constant and pre-determined by the modeler (as in the constant penalty model that will be discussed later). In contrast to other proxy-based models, the variable penalty model could provide a very good representation of the deprivation costs, as shown in Fig. 3. However, the variable penalty model is similar to its counterparts in not accounting for terminal deprivation costs. To compensate for such under-specification, equity constraints must be added. A variable penalty model that (1) correctly identifies deprivation time as the key variable, with parameters that closely replicate the deprivation cost function, and (2) properly considers the terminal costs, should be capable of producing solutions comparable to those produced by the social cost model.

4.2.2. Constant penalty models

These formulations define a threshold value, δ_p , for a relevant variable that triggers a penalty of magnitude ρ , as shown in Fig. 4. This provides the analyst with a parameter, the penalty, which could be used to tweak the model to produce reasonable results. There are a number of variants, including shortage costs/penalties

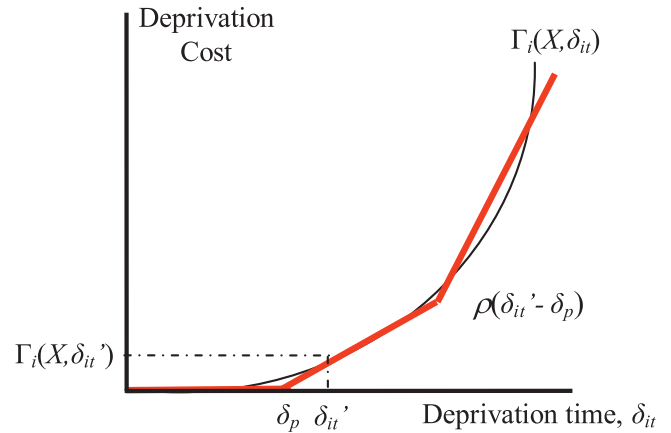


Fig. 3. Variable penalty and deprivation costs.

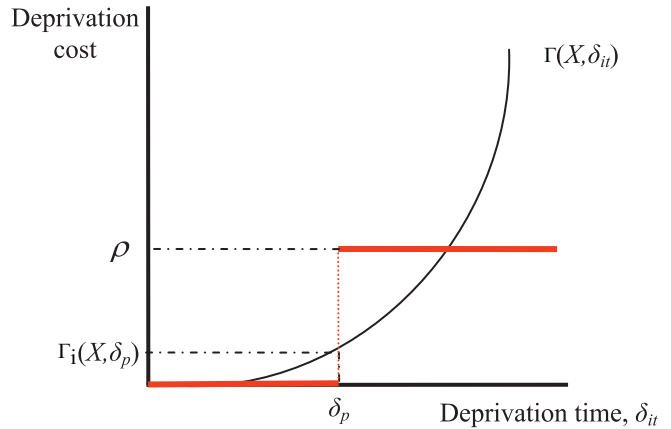


Fig. 4. Constant penalty and deprivation costs.

(Barbarosoglu and Arda, 2004; Yi and Kumar, 2007; Yi and Özdamar, 2007), cost on arcs representing the flow of “pseudo supplies” for unmet demand (Rawls and Turnquist, 2010), and penalty cost for a fraction of unmet demand of different commodities (Balcik et al., 2008), among others. These approaches are relaxations of the hard constraint models discussed later in the paper.

However, the constant penalty model has limitations in terms of its ability to correctly represent deprivation costs. It is simply not realistic to expect that a single constant penalty could account for such a wide range of values of deprivation costs. Fig. 4 shows that the threshold value, δ_p , is implicitly associated with a deprivation cost $\Gamma_i(X, \delta_p)$. Thus, for deprivation times just lower than δ_p the deprivation costs would be underestimated, and for deprivation times just higher than δ_p the deprivation costs would be overestimated, and then underestimated again once the actual deprivation cost $\Gamma_i(X, \delta_p)$ is higher than the penalty. The constant penalty model cannot capture the non-linearity of deprivation costs either.

The objective function for the constant penalty model can be expressed as:

$$\Psi_T^{CP}(X, T) = \Omega_T(X, T) + \sum_{i=1}^N \sum_{j=1}^J \rho y_{it_j} \quad (7)$$

$$\Psi_T^{CP}(X, T) = \Omega_T(X, T) + P_T(X, T) \quad (8)$$

where $P_T(X, T)$, total penalties incurred in implementing X during planning horizon T ; ρ , penalty incurred each time the defined level of service is not satisfied; y_{it_j} , Binary variable equal to 1 if the defined level of service is not met, 0 otherwise; δ_p , threshold value to trigger the penalty.

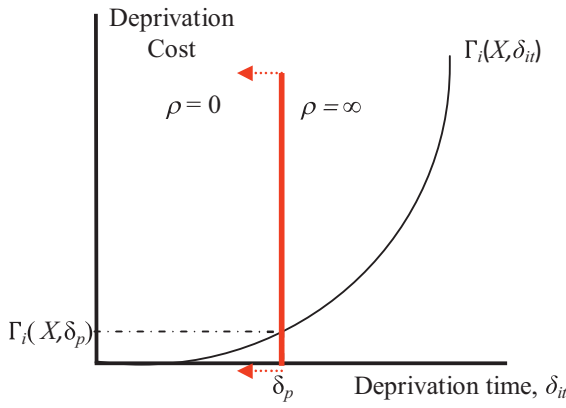


Fig. 5. Hard constraints and deprivation costs.

As shown in Eq. (7), although the constant penalty incorporates a term to represent the penalties accrued over time, the quality of the solutions depends on how accurately the penalty term captures those inter-temporal effects, i.e., the second and third terms of Eq. (5). Similar to the variable penalty case, the penalty is accrued by all nodes i for which the defined level of service is not met when evaluating at every delivery time t_j . In the case of the social cost model, the delivery patterns produced would depend on how the deprivation costs evolve over time, at all nodes in the system. This stands in contrast with the preset values of δ_p imposed by the penalty-based model, in which the trigger variable is the only factor. The expectation is that the constant penalty model cannot produce results of the same quality as those obtained from the social costs model.

4.2.3. Hard constraints

These are special cases of penalty-based models that incorporate hard constraints, such as “equity constraints,” to try to ensure that a commercial logistic model produces reasonable results. A fairly wide range of constraints is used, including: demand satisfaction before deadlines (Han et al., 2007), and flow conservation at nodes where amount supplied equals amount demanded (Rawls and Turnquist, 2010), among others. Mathematically, these models are equivalent to a constrained minimization of logistic costs, as shown in Eq. (9):

$$\Psi_T^{HC}(X, T) = \Omega_T(X, T) : \delta_{it_j} \leq \delta_p \quad (9)$$

Mathematically, the social cost and hard constraint models are very different. Most notably, the hard constraint model relies on constraints to capture the human suffering that is represented by the second and third terms of Eq. (5). Obviously, as illustrated in Fig. 5, no constraints can accomplish that feat. The hard constraint is equivalent to a penalty model, with a penalty $\rho = \infty$ for solutions exceeding threshold δ_p . The first problem with this approach is that the overly simplistic nature of the constraints—which could take two extreme values—does not capture the non-linear character of the deprivation costs. Also, since the constraints penalize solutions on the current time period, the model cannot consider the inter-temporal effects that determine the opportunity cost of the delivery strategy. It is unlikely that this model could lead to socially optimal solutions.

The constraints tend to lead to infeasible solutions, which further limits the model. The constraints implicitly cap the deprivation cost at $\Gamma_i(X, \delta_p)$ without considering whether such level of service is possible given the shortages in both critical supplies and the capacity to deliver them. The ensuing infeasible solutions force modelers to change the values of the constraints until seemingly reasonable results can be obtained. As a result, modeling becomes increasingly arbitrary and subjective.

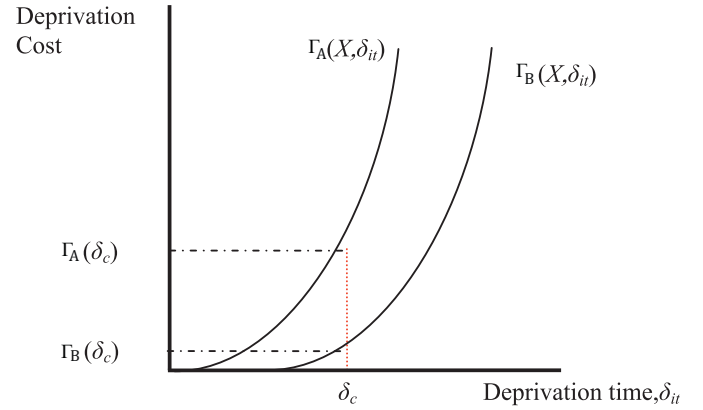


Fig. 6. Unmet demands and deprivation costs.

The key insight is that all penalty-based models have to correctly compute the deprivation costs at the various demand nodes, and properly consider the opportunity and terminal costs. The importance of these considerations is clearly illustrated with the numerical experiments in Section 5.

4.3. Minimization of unmet demands

The minimization of unmet demands has also been used in PD-HL modeling. The objective is to minimize the difference between the ideal delivery volumes based on predefined levels of service, and the actual amounts of supplies delivered to the demand nodes. The unsatisfied demand U_{it} at demand node i and time t is defined as follows:

$$U_{it_j} = q_{it_j} - d_{it_j} \quad (10)$$

where q_{it_j} is the amount of commodity demanded at node i at time t , and d_{it_j} is the amount delivered by X .

The objective function minimizes the sum of total unmet demand at all nodes and time periods:

$$U^* = \sum_{i=1}^N \sum_{j=1}^J U_{it_j} \quad (11)$$

Models minimizing unmet demands do not take into account any of the cost components—private or external—associated with the delivery strategy, as implied by Eq. (11). Having identified an optimal solution, the analyst has to compute delivery costs and routing if these were not incorporated in the model. It is difficult to compare this type of model to the social cost model given that it does not consider any of the cost components that the social cost model considers. One can assume that the unmet demand model would not produce delivery strategies with reasonable social costs given that it ignores all costs.

To illustrate the model's chief limitation, consider the situation in Fig. 6 showing the deprivation cost functions for two demand nodes, A and B. Assume that node A, the smaller of the two, has a longer deprivation time than node B; and that as a result of the longer deprivation time, the total unmet demand for A is equal to the total unmet demand for node B (this could also happen, for instance, if there was a larger initial stock of supplies at B). In this context, an optimization model minimizing unsatisfied demands would be insensitive to the beneficiaries' deprivation costs, as the unmet demands are equal. Therefore, if the solution calls for a delivery at time δ_c , a responder aiming to minimize unmet demands will optimize its objective by delivering the largest amount of supplies irrespective of the deprivation costs. However, as implied in the

figure, the total costs to individuals would be minimized by delivering first to node A, as it has a higher deprivation cost.

The unmet demand model also assumes additive demands at the beneficiary nodes, i.e., that the population will be able to consume the total unmet demand when a delivery arrives. However, a key characteristic of human deprivation is that demands are not additive, essentially invalidating the objective pursued in this formulation.

4.4. Discussion

The wide spectrum of PD-HL analytical formulations could be exemplified by five model types, and their corresponding objective functions, as summarized in Table 2. As shown, the objective functions of the five types are very different. While most models account for logistic costs in some fashion (the exception being the unmet demand model), they consider the impacts on human suffering in radically different ways. In the simplest form (the hard constraint model) the constraints impose a minimum level of service without considering its feasibility; in some models a constant penalty is added that cannot account for the non-linear nature of deprivation costs; while in others, a proxy for human suffering, i.e., unmet demands, is used that cannot differentiate among demand nodes that have equal unmet demands but different deprivation times. The only formulation that is structurally capable of capturing these variations—though with modifications—is the variable penalty model.

All of the models that based on proxy approaches fail to consider the terminal costs at the end of the planning horizon (the third term in the social cost model), which is absolutely necessary to properly account for inter-temporal effects. Simply adding another term to the objective function will not address this limitation, as some of the model structures do not lend themselves to the incorporation of such terminal costs. The objective functions of the hard constraints and unmet demand models would have to be holistically transformed for them to be able to incorporate terminal costs. The constant penalty model cannot consider terminal costs either, because the penalties are assessed in an all-or-nothing manner, once the trigger threshold is reached. Only the variable penalty model is potentially capable of considering terminal costs by adding a term that computes the penalties accrued from the last delivery to the nodes, to the end of the planning horizon. Thus, the variable penalty model could indeed produce solutions comparable to those of the social cost model, provided that the penalty function accurately represents deprivation costs, and that terminal costs are accounted for. The enhanced version of the variable penalty model capable of achieving this would be:

$$\Psi_T^{VP}(X, T) = \Omega_T(X, T) + \sum_{i=1}^N \sum_{j=1}^J \rho'(\delta'_{it_j} - \delta_p) y'_{it_j} + \sum_{i=1}^N \Gamma_i(X, T) \quad (12)$$

While more refined proxy-based measures could overcome the limitations discussed, one must question their utility. Their nature requires complex solution algorithms, and they do not overcome the fundamental criticisms of other proxy-based approaches (e.g., an inability to capture the non-linearity of deprivation costs, arbitrariness of key parameters). In this context, the authors' position is that using social costs as the objective function of PD-HL models is the best tradeoff between computational performance and quality of solutions.

5. Numerical experiments

In this section, numerical experiments are used to illustrate the implications of using proxy-based approaches. The experiments

are based on relatively simple and small distribution networks, as these provide insight into the comparative performance of the different approaches, without adding overwhelming complexity. The scenarios assume a unit travel cost of \$55/h (Holguín-Veras and Brom, 2008), and a deprivation cost function estimated with data collected about willingness to pay for water (Fig. 7). The limit value of the deprivation cost function—the point at which an individual dies after five days of deprivation—was assumed to be five million dollars (Weisbrod et al., 2009). The scenarios assume that loading/unloading time and costs could be disregarded.

The analyses show: (1) how, on the PD-HL system, the balance between demand and ability to deliver critical supplies determines the level of service that can be attained; and, (2) the comparative performances of the alternative formulations discussed in Section 4. The models are based on a basic set of assumptions made consistent with each other, to ensure comparability. In the case of the hard constraint model, it is assumed that a delivery must be made within five days of the last delivery, to prevent people from dying. The constant penalty and unmet demand models consider a \$10,000 penalty each time the desired level of service is not met (defined as one delivery per day to each node), or a demand is not satisfied. Two cases of variable penalty model are considered (with two- and three- piece-wise linear segments), and both approximations were estimated by minimizing the square errors with respect to the original Eq. (13).

The original equation is described by:

$$\gamma_g(\delta_{it}) = e^{(1.5031 + 0.1172\delta_{it})} - e^{(1.5031)} \quad (13)$$

Two- piece-wise linear approximation:

$$\gamma_g(\delta_{it}) = \begin{cases} 2393.9\delta_{it} & \text{if } 0 < \delta_{it} \leq 105 \\ 333642(\delta_{it} - 105) + 251359.5 & \text{otherwise} \end{cases} \quad (14)$$

Three- piece-wise linear approximation:

$$\gamma_g(\delta_{it}) = \begin{cases} 341.32\delta_{it} & \text{if } 0 < \delta_{it} \leq 85 \\ 39031(\delta_{it} - 85) + 29012.2 & \text{if } 85 < \delta_{it} \leq 105 \\ 282109(\delta_{it} - 105) + 809632.2 & \text{otherwise} \end{cases} \quad (15)$$

5.1. Experimental setup I: deprivation times vs. demand and supply

This analysis shows the interconnections between the ability of the PD-HL system to deliver critical supplies, population demands, deprivation times and deprivation costs. The first case assumes that a distribution center (DC) delivers water to a single demand node using a truck that works 24 h a day. Obviously, since there is only one demand node, the optimal policy is to deliver to it as much, and as frequently, as possible. Different values of cycle time (round trip from the DC to the node and back), and truck capacity (expressed as a percentage of the demand at the node) are considered. The costs are calculated for operations with 20 deliveries, plus the deliveries required to satisfy unmet demands.

The results in Table 3 panel (a) show the non-linear nature of the problem. As expected, the deprivation costs decrease with increases in truck capacity, and increase with cycle time. This reflects the (opposite) effects that these variables have on system throughput. Panel (b) shows that the average deprivation time increases with the cycle time, as it takes longer for the truck to deliver the supplies. These results confirm one of the paper's basic postulates: in conditions of scarcity, the optimal level of service that can be attained is much lower than what could be attained in

Table 2
Summary of objective functions.

Model class	Objective functions
Social cost model: Sum of logistic and deprivation costs, all nodes and time epochs	$\Psi_T^{SC}(X, T) = \Omega_T(X, T) + \sum_{j=1}^J \Gamma_{ij}(X, t_j) + \sum_{i=1}^N \Gamma_i(X, T)$
Variable penalty model: Sum of logistic costs and variable penalties, all nodes and time epochs	$\Psi_T^{VP}(X, T) = \Omega_T(X, T) + \sum_{i=1}^N \sum_{j=1}^J \rho'(\delta_{it_j} - \delta_p) y_{it_j}$
Constant penalty model: Sum of logistic costs and penalties, all nodes and time epochs	$\Psi_T^{CP}(X, T) = \Omega_T(X, T) + \sum_{i=1}^N \sum_{j=1}^J \rho y_{it_j}$
Hard constraint model: Constrained summation of logistic costs	$\Psi_T^{HC}(X, T) = \Omega_T(X, T) : \delta_{it_j} \leq \delta_p$
Unmet demand model: Minimize total unsatisfied demand	$U^* = \sum_{i=1}^N \sum_{j=1}^J U_{it_j}$

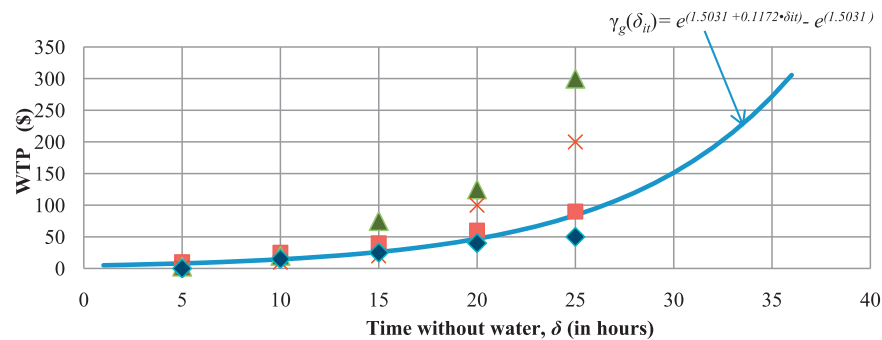


Fig. 7. Willingness to pay for water.

conditions of plenty. The results highlight the perils of imposing an unrealistic value of the threshold used in equity constraints, as it is impossible to know if such a threshold is attainable given the availability of resources and the uncertainty of demands.

5.2. Experimental setup II: comparative performance analyses

In these numerical experiments, the same problem was solved using different models to illustrate the comparative performance of the alternative approaches described in Section 4. Then, the corresponding social costs were computed to gain insight into the models' ability to reach socially optimal solutions. The experiments consider a relief agency that has a single vehicle to distribute full-truck loads of water to five nodes that each need one full-truck load of water per day. Two scenarios are considered: Scenario A assumes that the travel time is the same (6 h) between the DC and each of the five nodes; Scenario B assumes that the travel times between the DC and the nodes are 2, 5, 7, 11 and 13 h, respectively. The planning

horizon is 210 h. As before, there are no split deliveries, and the truck is required to return to the DC to reload after each delivery (these return trips are not listed in the sequences of trips). A schematic is shown in Fig. 8.

The scenarios are solved using relatively simple heuristics consistent with the basic postulates of each modeling approach. For the social cost formulation and its linear (two- and three- piece-wise linear) approximations, two heuristics were used: (1) delivering to the node with the highest marginal-benefit-to-marginal-cost ratio; and (2) delivering to the node with the highest marginal benefit. The marginal benefit is the reduction in the deprivation cost associated with delivering to the node; while the marginal cost is the sum of the increase in logistic and deprivation costs at all other nodes. In the hard constraint model, the heuristic searches for the solution that minimizes logistic costs subject to the constraint imposed. In the case of the constant penalty models, the objective is to find the distribution strategy that minimizes the sum of the logistic costs and the penalties for unsatisfied demands (a penalty

Table 3
Deprivation costs and average deprivation times for case I.

Truck capacity (% of demand)	Cycle time (days)			
	0.25	0.5	0.75	1
<i>(a) Deprivation costs (\$)</i>				
25	380.38	6133.97	100,881.67	1669,373.78
50	184.00	734.80	2965.06	12,022.32
75	181.48	546.29	1841.24	6732.05
100	178.97	357.78	717.42	1441.78
<i>(b) Average deprivation times (days)</i>				
25	0.91	1.83	2.74	3.65
50	0.48	0.95	1.43	1.90
75	0.37	0.74	1.11	1.48
100	0.24	0.49	0.73	0.98

Note: The average deprivation times include the transition period at the beginning of the operations.

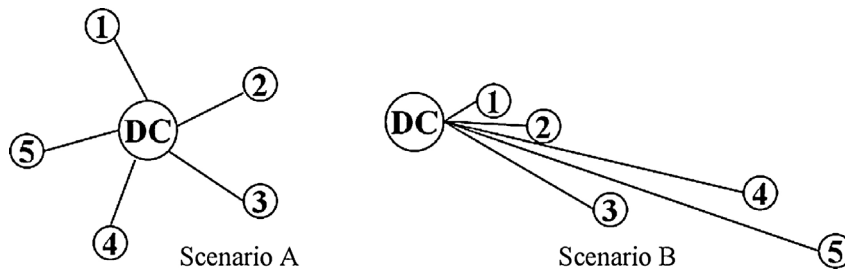


Fig. 8. Schematic of case II's scenarios.

of \$10,000 is charged each time daily demands at the nodes are not met). An equity constraint was also included to require that each node be served at least once every five days. The objective for the unmet demand model is to minimize the total unsatisfied demand per day among all nodes in the system. The solutions to the unmet demands and the penalty-based formulations were found by explicit enumeration.

In the first scenario, since each node has the same deprivation cost function and the same travel time to/from the depot, the approaches find multiple solutions with radically different results. The social cost formulation leads to stationary solutions in which the nodes are visited in a sequence that repeats time after time, as this is the best way to ensure a fair distribution of the resources. In the case of the penalty model, since what matters is to minimize the logistic-plus-penalty costs, in conditions where a penalty is to be assessed anyway, the model will favor delivering to the closest nodes. At some point, as the deprivation time reaches five days, a delivery is made to the node to satisfy the equity constraint. The penalty-based formulation is not likely to take into account that deprivation costs will increase significantly if some nodes are not serviced. In the case of the unmet demands model, the solutions obtained may provide even lower levels of service than those reached with the penalty-based formulation. As deprivation times are not considered in the unmet demand model, deprivation costs are not accounted for, as long as the unmet demand is minimized.

A comparison of the deprivation costs associated with the proxy approaches illustrates their differences. As shown in Table 4, the social cost formulation leads to a stationary (cyclic) pattern of distribution (1-2-3-4-5, 1-2-3-4-5, 1-2-3-4-5, 1-2), and a total deprivation cost of \$71,140. In contrast, the solution obtained by the (constant) penalty-based formulation ranges from \$71,140 when the optimal sequence (1-2-3-4-5, 1-2-3-4-5, 1-2-3-4-5, 1-2) is reached by chance, to \$4.3 million when remote nodes are served just in time to avoid penalties (e.g., 1-2-3-4-5-1-2-1-2, 1-2-3-4-5-1-2-1). In its worst case, the model minimizing unmet demands reaches a total deprivation cost of \$15 million, in the extreme case in which all deliveries are made to two nodes only, and beneficiaries at the other three demand nodes eventually die (e.g., 1-2, 1-2, 1-2, 1-2, 1-2, 1-2, 1-2, 1).

For Scenario B, as shown in Table 5, the penalty-based and unmet demand models result in shorter and more frequent delivery trips (19 and 24 trips, respectively) than the social cost model (14 trips). Travel costs were about the same for all strategies, as the truck was in continuous operation, and loading/unloading times and costs are disregarded. However, in terms of the social costs, the solutions are significantly different. The social costs for the penalty-based strategy are over 20 times higher (\$6.8 million vs. \$0.326 million) those for the social cost model strategy. The same holds true for the model minimizing unmet demands (\$10 million vs. \$0.326 million), as the maximum deprivation times are exceeded at demand nodes 4 and 5. The fundamental insight is that by not properly considering deprivation costs, proxy-based models will likely promote delivery strategies that fail to mitigate human suffering.

Table 6 shows the results for the social cost function, and its piece-wise linear approximations. It is important to acknowledge that the simple cases considered in the numerical experiments do not provide a solid platform to assess the performance of the variable penalty model. Since all demand nodes are equal, pretty much any cyclic pattern of distribution leads to an optimal, or quasi-optimal solution. This favors the variable penalty model, because if one uses the heuristic based on benefits, it leads to such solutions. In most complex cases, however, the results could be very different. Although only extensive experiments with a wide range of test problems provide certainty of the patterns of solutions, the authors are confident that these conjectures are robust.

In terms of numerical results, the delivery sequence estimated by the marginal-benefit-to-marginal-cost ratio-based heuristic (see Table 5), has an associated total deprivation cost that is less (\$325,592) than the one associated with the sequence found using the heuristic based on total benefits (\$338,618). However, the estimated results using the heuristic based solely on marginal benefits are consistently better than the results produced by the ratio marginal-benefit-to-marginal-cost heuristic, for the linear approximation models. Because the ratio does not identify any difference between deliveries to the nodes when the penalty increases in a constant rate, the ratio cannot exploit the non-linearity of the cost function, and produces suboptimal results. However, when comparing the results produced by the marginal-benefit-to-marginal-cost ratio, the social costs do improve when the approximation better resembles the deprivation cost function. In fact, the social cost ranges from \$5.8 to \$1.0 million in the two- and three-piece-wise linear approximations, respectively.

The numerical experiments conducted up to this point use a value of life typically used in the United States, \$5 million. The following analyses assess the impacts of different value of life assumptions (which impacts the deprivation cost function). The authors assumed a range of values of life, re-estimated the corresponding deprivation cost functions, and obtained the optimal delivery patterns. As Table 7 shows, if the value of life is relatively high (\$500,000 and above), the delivery strategy attempts to distribute equitably the supplies available. For values of life lower than \$500,000, the solutions start to favor delivery to the closest node (#1) since doing so reduces the total cost. Ultimately, if the value of life is very low, the solution reached corresponds to minimization of logistic costs.

However, a close inspection of Table 7 reveals that the use of averages introduces distortions and inconsistencies, like those affecting the statistics for node 4. For that reason, the authors decided to create Table 8 that shows the maximum delivery times. The results clearly show that if the value of life plays a significant role in the social costs, the model produces a delivery strategy that leads to all nodes having the same value of maximum deprivation time. In essence, the model minimizes total social costs by socializing the level of suffering. In contrast, as the value of life becomes smaller its role in the social cost model increasingly vanishes. This leads to delivery strategies that favor the closest nodes

Table 4
Heuristics solutions for scenario A.

Strategy	Objective function		
	Social cost formulation Deliver next to the node with maximum: (marginal benefit/marginal cost)	Penalty based formulation Minimize total cost (i.e., travel + penalty), deliver at least once every 5 days	Unmet demands Arrange deliveries to minimize total unmet demands per day
Delivery pattern (returning to depot after each node)	1-2-3-4-5-1-2-3-4-5-1-2-3-4-5-1-2	1-2-3-4-5-1-2-1-2-1-2-3-4-5-1-2-1	1-2-1-2-1-2-1-2-1-2-1-2-1-2-1-2-1
Number of deliveries	17	17	17
Travel costs	\$11,220	\$11,220	\$11,220
Penalties for unmet demands	–	\$280,000	\$280,000
Total deprivation costs	\$71,140	\$4,297,032	\$15,001,178
Nodes exceeding max deprivation times	0	0	3

Table 5
Heuristic solutions for scenario B.

Strategy	Objective function		
	Social cost formulation Deliver next to the node with maximum: (marginal benefit/marginal cost)	Constant penalty based formulation Minimize total cost (i.e., travel + penalty), deliver at least once every 5 days	Unmet demands Arrange deliveries to minimize total unmet demands per day
Delivery pattern (returning to depot after each node)	5-1-2-3-4-5-1-2-3-4-5-1-2-3	1-2-3-1-2-3-1-2-4-5-1-2-3-1-2-3-1-2-4	1-2-3-1-2-3-1-2-1-2-3-1-2-3-1-2-3-1-2-3-1-2
Number of deliveries	14	19	24
Travel costs	\$11,330	\$11,550	\$11,550
Penalties for unmet demands	–	\$250,000	\$210,000
Total deprivation costs	\$325,59	\$6,807,77	\$10,003,154
Nodes exceeding max deprivation times	0	0	2

Table 6
Solutions for scenario B from different heuristics.

Deliver next to the node with maximum	Objective function				
	Social cost formulation	Variable penalty model			
		2 Piece linear approximation		3 Piece linear approximation	
		Benefits	Marginal benefit/marginal cost	Benefits	Marginal benefit/marginal cost
Delivery pattern (returning to depot after each node)	5-4-3-2-1-5-4-3-2-1-5-4	5-4-3-2-1-5-4-3-2-1-5-4	1-1-2-1-3-1-2-1-4-1-3-2-5-1-2-3-1-4-1-2	5-4-3-2-1-5-4-3-2-1-5-4	1-1-2-1-3-1-2-1-4-1-5-1-3-2-1-4-1-2-5-3
Number of deliveries	12	12	20	12	19
Travel costs	\$11,000	\$11,000	\$11,000	\$11,000	\$11,770
Penalties for unmet demands	–	–	–	–	–
Total deprivation costs	\$338,618	\$338,618	\$5,758,786	\$338,618	\$1,090,080
Nodes exceeding max deprivation times	0	0	0	0	0

which enjoy much lower deprivation times, as doing so leads to minimization of the costs. In contrast, the nodes that are farther away experience drastic increase in the deprivation times. These results are consistent with the observation made earlier in the paper that underestimating the importance of deprivation costs leads to solutions that resemble logistic cost minimization.

The deprivation costs for the social cost and constant penalty models are illustrated in Fig. 9, which shows how the temporal distribution of deprivation costs per node at each time epoch (delivery time) leads to large social costs. In the constant penalty case, since there is no concern for deprivation costs, deliveries to distant nodes (nodes 4 and 5 in the example) are delayed as much as possible

Table 7
Average delivery times per node vs. valuations of human life.

Human life valuation	Delivery pattern	Total of deliveries	Total deliveries (avg dep time) per node				
			Node 1	Node 2	Node 3	Node 4	Node 5
5,000,000	5-1-2-3-4-5-1-2-3-4-5-1-2-3	14	3(60.0)	3(62.3)	3(62.3)	2(70.5)	3(55)
500,000	5-1-2-3-4-5-1-2-3-4-5-1-2-3	14	3(60.0)	3(62.3)	3(66.3)	2(70.5)	3(55)
50,000	5-1-2-3-4-1-5-2-3-1-4-5-2-1-	5	4(48.5)	3(62.3)	3(66.7)	2(72.5)	3(56.3)
5000	5-1-2-3-4-1-5-2-3-1-4-5-2-1-	5	4(48.5)	3(62.3)	3(66.7)	2(72.5)	3(56.3)
500	5-1-2-3-4-1-5-2-3-1-4-5-2-1-	5	4(48.5)	3(62.3)	3(66.7)	2(72.5)	3(56.3)
50	5-1-2-3-1-4-2-1-5-3-1-2-4-1-	16	5(34.4)	3(47.7)	3(69.0)	2(79.5)	3(62.3)

Note: Node 4 was in line to receive a delivery that did not take place because the planning horizon ended. This missing delivery impacts the node's number of deliveries and average deprivation time.

Table 8
Maximum deprivation time per node vs. valuations of human life.

Human life valuation	Delivery pattern	Total of deliveries	Maximum deprivation time (h)				
			Node 1	Node 2	Node 3	Node 4	Node 5
5,000,000	5-1-2-3-4-5-1-2-3-4-5-1-2-3	14	76	76	76	76	76
500,000	5-1-2-3-4-5-1-2-3-4-5-1-2-3	14	76	76	76	76	76
50,000	5-1-2-3-4-1-5-2-3-1-4-5-2-1-	5	62	76	50	50	50
5000	5-1-2-3-4-1-5-2-3-1-4-5-2-1-	5	62	76	5,0	5,0	5,0
500	5-1-2-3-4-1-5-2-3-1-4-5-2-1-	5	62	76	500	500	500
50	5-1-2-3-1-4-2-1-5-3-1-2-4-1-	16	44	58/76*	50	50	94

*Note: The maximum delivery time in terms of completed deliveries is 58 h. However, if one considers the delivery time for what would be the last delivery that could not be delivered because of the planning horizon, the maximum delivery time increases to 76 h.

(Fig. 9b). Under general assumptions, the models based on proxy measures tend to provide a poor level of service to distant nodes, which translates into high deprivation costs. Furthermore, since the solution for the constant penalty-based formulation is independent of the magnitude of the penalties chosen, increasing the magnitude of the penalty does not improve the quality of the solution. The example shows that the social cost formulation provides a uniform level of service to all nodes (after an initial period of stabilization), incurring much lower deprivation costs (Fig. 9a). The fact is that, in situations of scarcity, the optimal policy is to distribute the suffering equitably so that no one accrues extremely high deprivation costs.

6. Impacts of estimation errors

This section investigates how errors in the estimation of the deprivation cost function could impact the quality of the solutions. This topic is of theoretical and practical significance because of the inherent challenges in estimating deprivation cost functions. There are valid reasons to expect errors in the estimation of deprivation costs: there is no market to assign a value, there are measurement errors, instrumental and proxy variables may be needed, the model specification may be incomplete, and individual taste variations may be a factor (Manski, 1973). In essence, if the social cost formulation's merit is contingent on an accurate deprivation cost function,

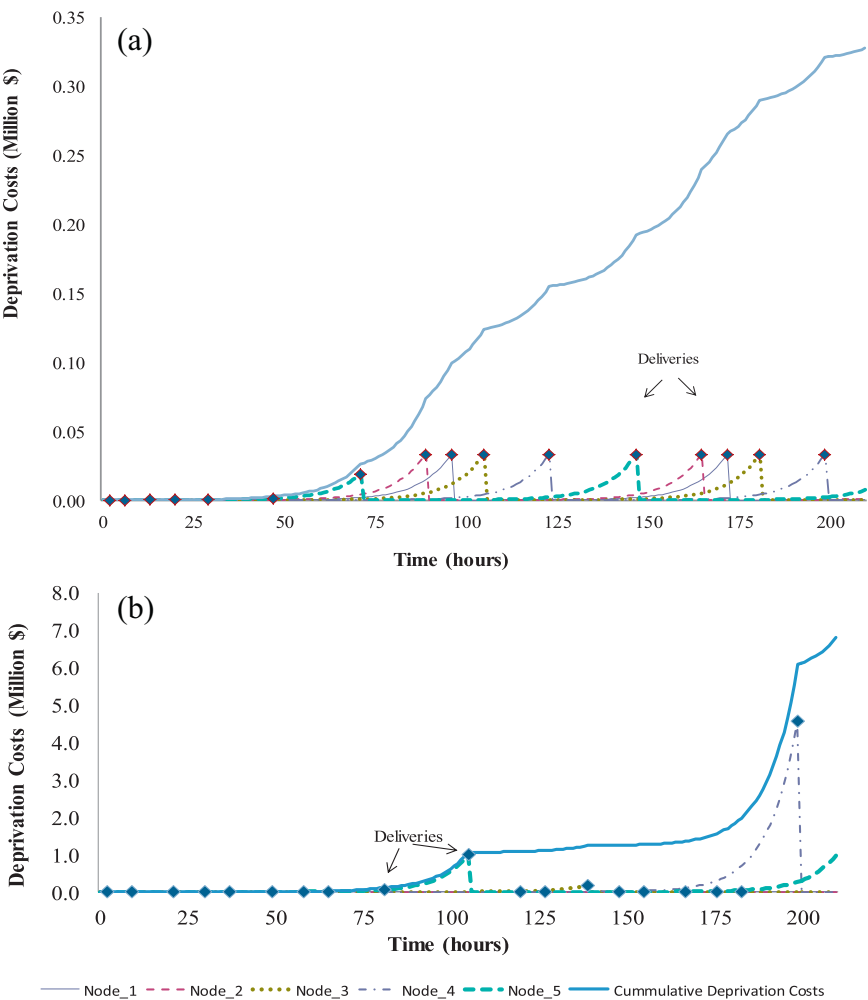


Fig. 9. Social costs for: (a) social cost model, and (b) constant penalty model.

which may not be available, using proxy measures could be justified. On the other hand, if the results are robust to errors in the estimated deprivation cost function, then the social cost formulation would clearly be the superior alternative.

To analyze the impacts of such errors, assume the existence of a single-valued monotonic convex function, $\gamma_g(\theta_g, \delta_{it})$, that represents the population's (true, though unknown) generic deprivation cost function. Consider the case in which, because of observational and measurement errors, the estimating deprivation cost function, i.e., $\hat{\gamma}_g(\hat{\theta}_g, \delta_{it})$ that is available to the analyst contains an unknown error. The total impact of the error on the delivery strategy also depends on the number of individuals impacted at the node. However, since the standard practice is to have points of distribution serving approximately the same number of individuals, one could focus on the deprivation cost functions, as these will be the main factors impacting the delivery strategy.

Whether the estimating function overestimates/underestimates the true deprivation cost function is an important insight when analyzing the role of the errors. Because of the assumptions of monotonicity and convexity, the number of possibilities is relatively small: (1) a function could overestimate/underestimate the other for a range of values, and then could underestimate/overestimate for the remainder; or, (2) a function could either overestimate/underestimate the other for the entire range. These two cases are referred to as partial dominance and strict dominance, respectively, and are shown in Fig. 10. Panel (a) shows two examples in which the estimating functions do not consistently overestimate/underestimate the true function (the solid lines); in panel (b) the estimating function consistently overestimates/underestimates the true one.

In the case shown in panel (a), the nature of the differences between the estimating and the true deprivation cost function could indeed impact the relative importance of (estimated) deprivation costs at the demand nodes and, consequently, how far off the resulting delivery strategy is from the optimal one. In these cases, the solutions will be determined by the estimating function used. However, the case shown in panel (b)—where the estimating function either dominates or is dominated by the true function—leads to important observations. When the estimating deprivation cost function dominates the true one:

$$\hat{\gamma}_g(\hat{\theta}_g, \theta_{it}) > \gamma_g(\theta_g, \delta_{it}) \quad \forall t \quad (16)$$

Conversely, if $\hat{\gamma}_g(\hat{\theta}_g, \delta_{it})$ is dominated by $\gamma_g(\theta_g, \delta_{it})$:

$$\gamma_g(\theta_g, \delta_{it}) > \hat{\gamma}_g(\hat{\theta}_g, \delta_{it}) \quad \forall t \quad (17)$$

These two cases, when looked at in detail, illustrate the quality of the results produced by the model based on social costs.

Case I: $\gamma_g(\theta_g, \delta_{it}) > \hat{\gamma}_g(\hat{\theta}_g, \delta_{it}) \quad \forall t$

Here, the estimating function used in the model underestimates the true function for the entire range of deprivation times. The deprivation costs therefore have less influence than they should in the decision. One could expect that increasing levels of underestimation of deprivation costs will lead to social cost functions that increasingly resemble logistic costs, and solutions that increasingly resemble those from logistic costs minimization. If the underestimation is extreme, the optimization problem solutions would be equal to those produced by minimization of logistic costs (with concomitant high social costs).

Case II: $\hat{\gamma}_g(\hat{\theta}_g, \delta_{it}) > \gamma_g(\theta_g, \delta_{it}) \quad \forall t$

The estimating function in Case II overestimates the true deprivation costs, implying that the deprivation costs are given more weight than they should have. However, under the prevailing conditions of most disasters, this error is likely to only have a minor

impact on the delivery patterns produced by the model. When deprivation costs dominate logistic costs (which is expected to be the case in PD-HL after large events) the delivery patterns are increasingly determined by the values of the deprivation costs for the different demand nodes. In this context, the relative levels of urgency at the nodes will determine the optimal delivery strategy under the assumption of monotonicity and strict dominance, an estimating deprivation cost function that overestimates the true deprivation costs will not change the rank order of nodes in terms of urgency (though the values of the social costs would be different). Thus, one could expect that the delivery patterns produced by the estimating function would be very similar to the ones obtained with the true deprivation cost function. This suggests that being accurate in the estimation of the deprivation cost function is of relatively secondary importance, as the errors in the estimation would have only a minor impact on the nature of the solutions.

Within a range of conditions optimal solutions can only be found with an accurate deprivation cost function. These cases are likely to arise when deprivation and logistic costs are of comparable order of magnitudes, where different estimated deprivation cost functions could lead to different solutions. However, since most disasters involve large shortages, one can expect that the deprivation costs would be much larger than logistic costs. Under these conditions, the paper's conjecture likely holds: robust solutions will be obtained using an estimating deprivation cost function that strictly dominates the true one.

The authors conducted a set of numerical experiments to assess the validity of this conjecture. Table 9 shows the results for a number of scenarios in which the parameters of the deprivation cost have been assumed to have different levels of error (i.e., −100%, −50%, −25%, +25%, +50%, and +100%) with respect to the true values, shown in Fig. 7. Each of the scenarios was solved using the estimating (and erroneous) deprivation cost function. Then the true values of the social costs were computed to assess how using the erroneous function impacts the true costs. It is important to note that, since the heuristic used does not search for a global optimum, it is always possible to find cases (e.g., −25% and +50%) that lead to lower social costs (by chance), even if the estimating deprivation cost is in error.

As expected, if the estimating function significantly underestimates the true deprivation costs, the solution resembles the one for commercial logistics that minimizes logistic costs, resulting in higher levels of service to the nodes closest to the DC. Because of the underestimation, the heuristic takes longer to reach a stationary solution (or may not reach one at all). In contrast, if the estimating function dominates the true function, the stationary solutions are the same, though the values of social costs produced by the model with the estimating deprivation cost function are erroneous. A comparison of the results suggests that, though the mean deprivation times are not significantly different, the social cost formulation leads to narrower intervals of values (34.4, 73.5), than those for the cases that underestimate the true social costs (25.1, 105.0). In the latter case, while some nodes are served much better than they warrant, other nodes are underserved.

The results confirm that as long as the estimating function dominates the true one, the delivery patterns are likely to be similar to those obtained using the true function, though the value of the social costs would be different. This is greatly significant since the ultimate goal is a delivery strategy that is optimal for the conditions in the field; the actual value of social costs is merely a mathematical accessory to compute that optimal delivery strategy. Assuming an estimating function that overestimates the true deprivation cost function may be all that is necessary to obtain the delivery pattern that would be derived from the true (and unknown) deprivation cost function. This reduces the need for an accurate estimation of the deprivation cost function, facilitates

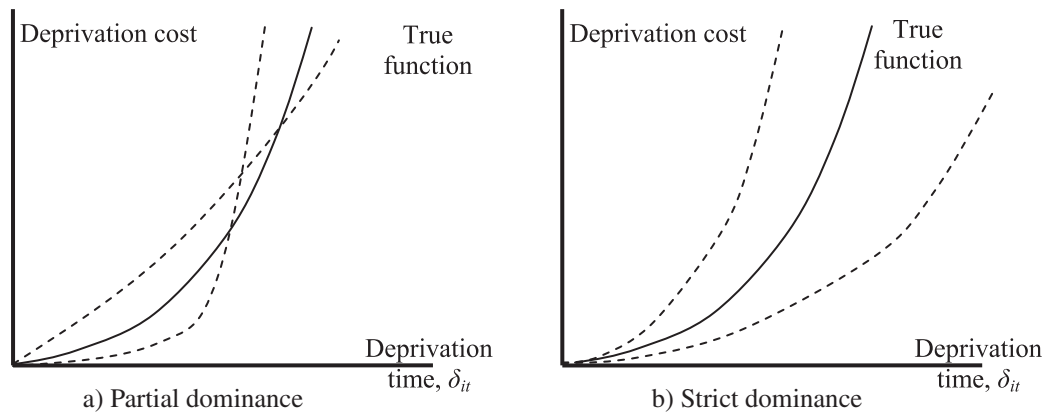


Fig. 10. Illustrative cases of estimation errors. Note: The estimating functions are shown in dashed lines.

Table 9
Solutions obtained for different levels of estimation error.

Error (%)	Delivery pattern	Deprivation costs (\$)			Deprivation times (h)			
		Estimated	True	Difference	Min	Max	Mean	Median
–100	1,2,3,1,2,3,1,2,4,5,1,2,3,1,2,3,1,2,4	–	6,807,878	–6,807,878	25.1	105.0	59.5	41.8
–50	1,1,1,1,2,1,3,4,5,1,2,3,4,1,5,2,3,1,4	4197	353,544	–349,347	24.8	79.5	58.4	63.0
–25	1,1,1,2,3,1,4,5,2,1,3,4,5,1,2,3,4	35,670	326,674	–291,003	27.5	75.5	58.0	61.7
0	1,1,2,1,3,4,5,2,1,3,4,5,2,1,3,4	327,712	327,712	0	34.4	73.5	57.9	60.3
+25	1,1,2,1,3,4,5,2,1,3,4,5,2,1,3,4	2,978,795	327,712	2,651,083	34.4	73.5	57.9	60.3
+50	1,1,2,3,1,4,5,2,1,3,4,5,2,1,3,4	27,550,490	320,842	27,229,649	34.4	73.5	57.9	60.3
+100	1,1,2,3,1,4,5,2,3,1,4,5,2,1,3,4	2,212,817,818	320,842	2,212,496,977	34.4	73.5	57.9	60.3

Note: The results for the –100% case correspond to the one for the logistic costs minimization.

implementation of the social cost formulation, and leads to delivery strategies that will be much better than those that could be obtained using other proxy measures of human suffering. These considerations are not trivial or esoteric; they have a basis in the real world and the present time. Consider, for instance, the increasing pressure on humanitarian organizations to apply commercial logistics principles and software, and the trend to outsource logistics to commercial or quasi-commercial organizations. Given the comparisons discussed, and their related impacts, such changes may lead to a higher emphasis on logistic cost minimization. If this is combined with poorly estimated deprivation costs, the consequences could be serious. A commercial firm could surely reduce delivery costs, but might do so by inadequately serving remote beneficiaries, causing huge deprivation costs. True, “equity constraints” could be incorporated to the model to prevent this from happening which will lead to the predictable results discussed in Section 4.2.2.

7. Conclusions

The paper argues for the importance of incorporating welfare economic concepts into PD-HL analytical models, and for the use of social costs as the objective function. After large disasters and catastrophes: the local markets that normally allow buyers and sellers to conduct economic transactions are imperiled or destroyed; local inventories maintained by households and businesses are also largely destroyed; the availability of currency and credit is limited; and private sector supply chains are severely curtailed or severed. Given a context in which the only practical alternative to reduce human suffering is for relief groups to distribute critical supplies for free, the relief effort’s impacts become economic externalities, not accounted for by economic transactions. As a result, proper estimation of economic welfare requires the use of social costs, or, the

summation of all private and external costs and benefits accrued by the economic agents involved.

The paper identifies three key components of economic welfare that must be computed at a given delivery epoch. The first is the economic impacts on the relief agency that incurs the logistic costs associated with the delivery of relief aid. The second is the reduction of the deprivation costs for the individuals who receive aid at that point in time. The third consideration is the opportunity cost of the delivery strategy, or the increase in deprivation costs for the population that does not receive supplies, and has to wait for future deliveries. Thus, identifying the optimal delivery strategy requires consideration of the inter-temporal externalities (both positive and negative) involved; this is the only way to assess the strategy’s full impacts. Single period models are therefore not appropriate for PD-HL modeling.

The formulation introduced in the paper uses a definition of social costs in its objective function that is multi-period in nature, enabling correct computation of the inter-temporal externalities. Second, the formulation does not require the specification of equity constraints, as the level of service that the PD-HL system could deliver is endogenously determined. These characteristics enable the accurate consideration of a delivery strategy’s impacts, without the analyst needing to specify equity constraints that—as amply discussed—will most likely be arbitrary and subjective in nature.

Proper consideration of these economic impacts is challenging, but essential. While logistic costs can be easily estimated using standard cost estimation techniques, the best way to capture a delivery strategy’s impacts on the population is through the formal estimation of a deprivation cost function. Such a function represents the economic cost of human suffering as a function of deprivation time, or, the amount of time that an individual is without a given good or service. Although such estimation is

based on economic valuation techniques, the authors' conclude that it is superior and preferable to using the proxy measures suggested in the literature. This invites the practical question of how to empirically estimate deprivation cost functions, which is an area where further research is needed. For the key supplies needed after disasters, by estimating deprivation cost functions the modeling community has an objective way to assess the impacts of delivery strategies.

A key difference between commercial and humanitarian logistics is central to the paper's postulates. In commercial logistics, delivery strategies are determined by efficiency and customer satisfaction considerations within an environment in which supply is equal or larger than demand. Humanitarian logistic operations attempt to minimize the social costs of a disaster's impacts, within a context of severe shortages. This requires making decisions about who will and who will not receive aid, at a given moment in time. PD-HL models inspired by the commercial logistic paradigm are ill-equipped to consider such complex tradeoffs. In most cases, the penalties are too crudely defined to represent the non-linearity of deprivation costs (only the variable penalty model could provide a meaningful depiction of these costs). Secondly, most proxy-based models fail to properly account for the opportunity costs associated with delivery decisions, which include the impacts of delivering to a node at a point in time, and the terminal costs that accrue from the last delivery to the end of the planning horizon.

While as a whole the proxy-based models are inadequate for PD-HL, they exhibit different levels of performance. The hard constraint and unmet demand models do not account for the impacts on deprivation costs in any meaningful way, nor do they consider terminal costs. The constant penalty model attempts to account for deprivation costs, but fails because the penalty cannot address the non-linearity of the deprivation cost function. Moreover, because the penalty is only assessed once the trigger is exceeded, the terminal costs are not properly accounted for. The variable penalty model could indeed accurately represent the deprivation cost function if the parameters are properly selected, and deprivation time is specified as the key independent variable. However, since the terminal costs are not accounted for, the variable penalty model is not likely to lead to socially optimal solutions. (Incorporating terminal costs in the variable penalty model would solve this problem.) Generally, the addition of equity constraints is nothing more than a reflection of the models' structural limitations.

The paper's analyses suggest that accuracy in the estimation of the deprivation cost function may be of secondary importance. As long as the model's estimating deprivation cost function strictly dominates the true function, the solutions will likely be similar. This result is expected to hold whenever the deprivation costs are much larger than the logistic costs, as will be the case in the immediate aftermath of large disasters. This insight, together with the others provided, chart a practical avenue for the robust estimation of delivery plans that minimize the human suffering produced by large disasters and catastrophic events.

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